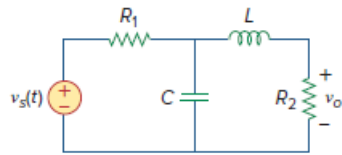


Problem

Using Fig. 10.95, design a problem to help other students understand source transformation.

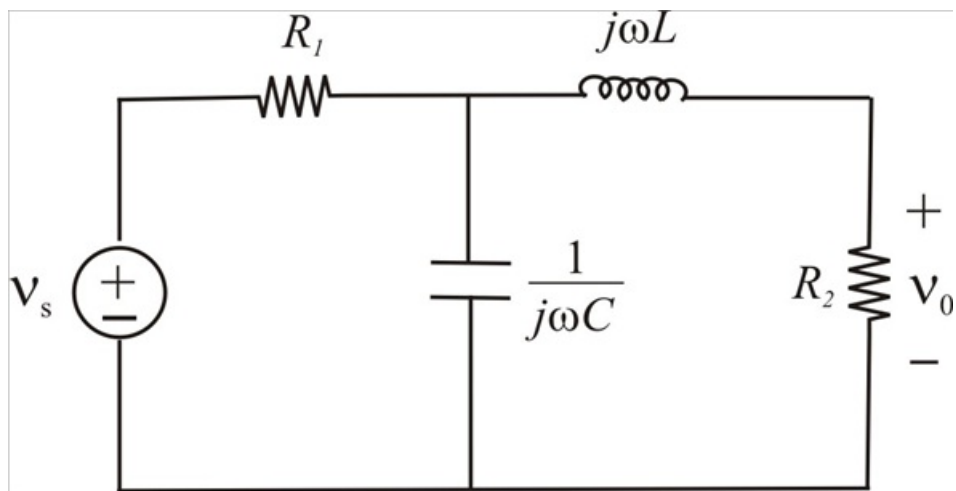
Figure 10.95:



Step-by-step solution

Step 1 of 6

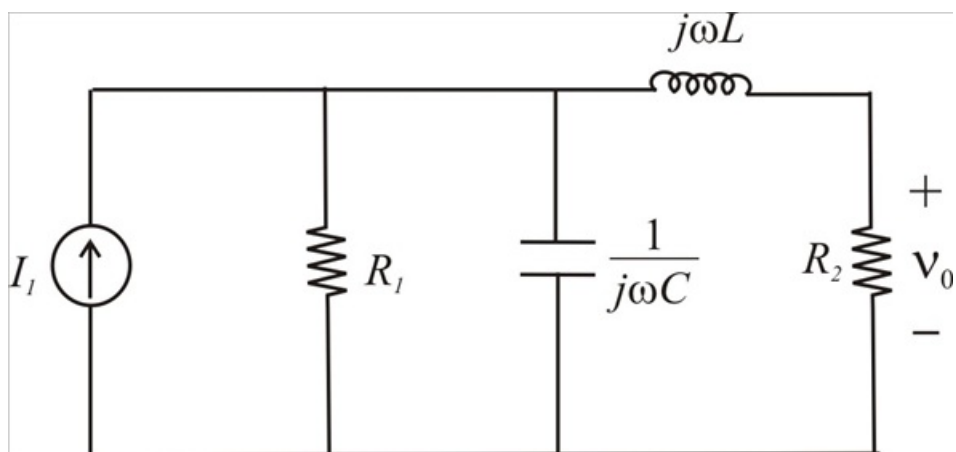
Given circuit can be represented as,



Step 2 of 6

We transform the voltage source to current source is shown below

$$I_1 = \frac{V_s}{R_1}$$



Step 3 of 6

From the above circuit,

The parallel combination of R_1 resistor and $\frac{1}{j\omega C}$ impedance is,

$$Z = R_1 \parallel \frac{1}{j\omega C}$$

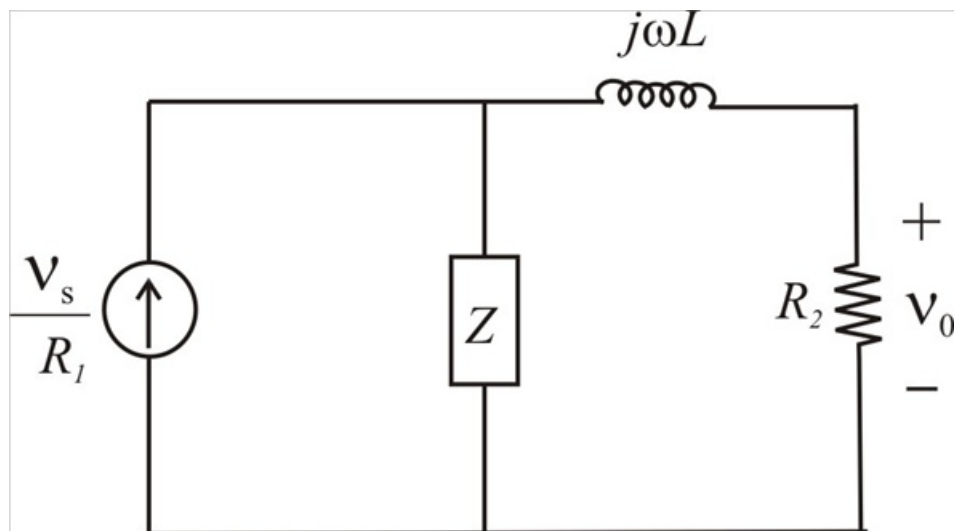
$$Z = \frac{R_1 \times \frac{1}{j\omega C}}{R_1 + \frac{1}{j\omega C}}$$

$$Z = \frac{\frac{R_1}{j\omega C}}{\frac{j\omega C R_1 + 1}{j\omega C}}$$

$$Z = \frac{R_1}{j\omega C R_1 + 1}$$

Step 4 of 6

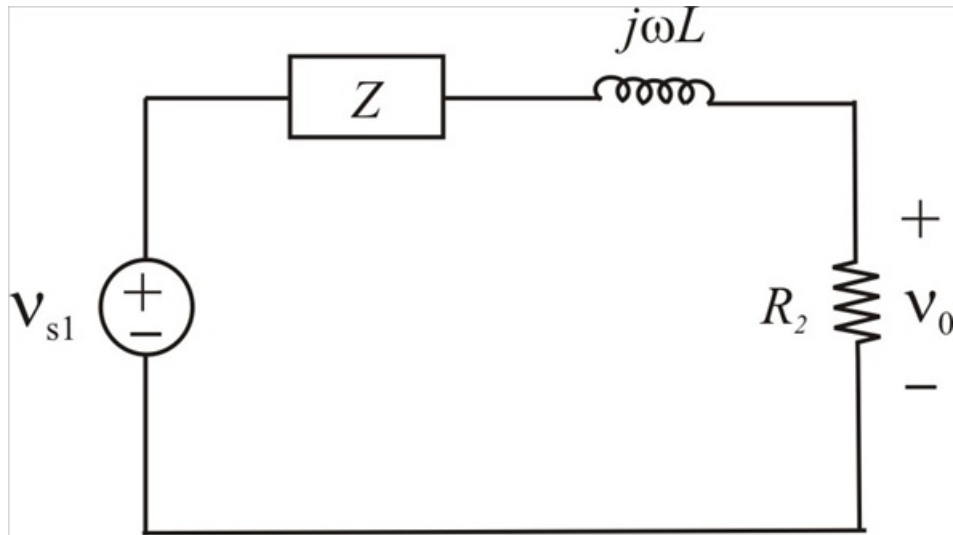
By redrawing above circuit,



Step 5 of 6

Convert current source to voltage source is shown below

$$v_{s_1} = \frac{v_s}{R_1} \times Z$$



Step 6 of 6

By using voltage division to above circuit.

$$v_0 = v_{s_1} \times \left[\frac{R_2}{R_2 + Z + j\omega L} \right]$$

$$\Rightarrow v_0 = \left(\frac{v_s}{R_1} \right) \times Z \times \left[\frac{R_2}{R_2 + \frac{R_1}{j\omega CR_1 + 1} + j\omega L} \right]$$

$$\Rightarrow v_0 = v_s \times \left[\frac{\left(\frac{R_1}{j\omega CR_1 + 1} \right) R_2}{R_1 \left(R_2 + \frac{R_1}{j\omega CR_1 + 1} + j\omega L \right)} \right]$$

$$\Rightarrow v_0 = v_s \times \left[\frac{R_1 R_2}{j\omega CR_1^2 R_2 + R_1 R_2 + R_1^2 + (j\omega)^2 CL R_1^2 + j\omega L R_1} \right]$$

$$\therefore v_0 = v_s \times \left[\frac{R_1 R_2}{R_1 (R_1 + R_2) + j\omega CR_1^2 R_2 - \omega^2 CL R_1^2 + j\omega L R_1} \right]$$